

QoS-equilibrium slot allocation for beam hopping in broadband satellite communication systems

Han Han¹ · Xueqiang Zheng¹ · Qinfei Huang² · Yuan Lin¹

Published online: 2 April 2015

© Springer Science+Business Media New York 2015

Abstract In broadband satellite communication systems, beam hopping (BH) is more and more welcome to compensate the low-power-efficiency wide beam and the low-spectrum-efficiency multiple spot beams simultaneously. As a flexible coverage method, BH demands efficient hopping strategy to reach the desired quality of service (QoS). In this paper, we consider the QoS equilibrium, in particular the delay fairness, among different cells illuminated by one hopping beam through designing the hopping strategy on the condition of capacity satisfaction. Due to the physical limitation that hopping speed is much slower than packet arrival speed, the ideal first-come-first-serve packet scheduling is not realizable, which compels us to pursue the optimal slot scheduling instead. In this paper, we consider a long-term delay fairness problem by per-period slot allocation for BH, using instantaneous and statistical information. To deal with the complex stochastic programming problem, we introduce the time-sharing principle to uncouple the variables and adopt stochastic gradient theory to deduce the suboptimal close-form solution. Our contributions are in three folds: firstly, it is the first time to consider the delay fairness not the traditional capacity fairness for BH; secondly, a general per-period decision scheme is proposed for the unique slot allocation in the perspective of queuing; thirdly, a suboptimal close-form solution for the per-period slot allocation is obtained, which could also cope with burst traffic cases. In

simulation, we verify our method's advantage in terms of delay fairness comparing with the existing similar works.

Keywords Beam hopping · Stochastic gradient theory · Satellite communication · Delay

1 Introduction

In the satellite communication area, more and more traffic will emerge, in form of data, voice or stream, therefore requiring broadband communications. In order to increase the total satellite capacity, small beams such as spot beams are widely in use, with each illuminating a cell on the ground to serve a coverage area. As a result of the spatial separation, partial spectrum could be reuse in different cells simultaneously, so as to increase the spectrum efficiency. However, it may be impractical to apply full frequency reuse because of excessive co-channel interference, which is difficult to escape using existing interference mitigation techniques.

One possible solution of the full spectrum reutilization is to separate the cells illuminated by small beams in time domain (one beam illuminates only one cell at one time), which is called beam hopping (BH) [1–11]. Additional welfare of this technology is that less payload is needed to perform the same communication task, because less beams are needed. In [1], the beam hopping technology is investigated from the viewpoint of throughput, with the hopping pattern complying with the keep-out rule. But it is on a per-packet hopping basis, which is an ideal assumption. European Space Agency (ESA) has promoted a project of beam hopping techniques for multi-beam satellite systems [2] in 2004, and Hughes Network System has developed and launched the Spaceway-3 satellite system using the hopping

✉ Han Han
hhhd20042008@gmail.com

¹ Institute of Communications Engineering, PLA University of Science and Technology, Nanjing 210007, China

² Beijing Capital Highway Development Group Co. Ltd, Beijing 100073, China

spot beam [3]. Several United States Patents [4–6] are also on the topic, greatly promoting the related researches. Pecorella proposes an implementable system scheme of beam hopping for DVB-RCS in [7] while J. Anzalchi performs detailed system simulation and performance comparison with non-hopped systems in [8]. Jiang proposes two novel capacity optimization approaches considering the power and carrier allocation in [9], and shows that BH system turns out better than multi-beam non-orthogonal frequency reuse system especially for non-real time services. As for the beam hopping antennas, the author of [10] presents a more-realistic focal-array-fed reflector antenna framework based on a cell-reuse scheme. And recently, the beam hopping technology is proposed to combine with the cognitive radio technology to achieve higher spectral efficiency in future satellite system in [11].

For BH, the assignment of time slots to all the queuing cells is critical for gaining good performance. Let us denote the hopping decision period (defined as Beam Hopping Window latter) comprised of fixed equal-length slots, and assume the visiting time for each cell (all cells must be visited at least one time in a period) is integer multiples of slots. The existing studies of the time assignment methods could be classified into two categories according to its decision epoch, namely the per-slot assignments [12, 13] and the per-period assignments [9]. In [12], Neely et al. consider it a server allocation problem, and design a ‘Choose-the-K-Largest-Connected-Queue’ policy with the stabilization goal. Choi and Chan [13] prove that cell with the highest attenuation-weighted demand should be served first per slot in order to balance the traffic demands and the channel capacity when the beam number is less than the cell number. In [9], the author carries out the time slot allocation following Fairness Cost Function and n th Order Difference Cost Function. As for its fairness criterion, the author matches the allocated capacity to the requested capacity, which is actually a capacity fairness criterion. Since the criterion is actually complying with the capacity allocation, it is stiff and losing sufficient consideration of QoS. On one hand, the capacity fairness is not equivalent to the QoS fairness. At least, the packet delay performance of different cells cannot be balanced by the fair capacity, which could be explained by the queuing theory [14] or the effective capacity theory [15]. On the other hand, the capacity fairness has not considered the traffic fluctuation. If the traffic is at its peak value, the allocated resource may be inadequate, while wasteful if the traffic is at its valley value. When multiple cells have such inadequate-wasteful troubles simultaneously, resources could be allocated smartly so as to make the QoS fairer.

We consider the delay fairness, not the capacity fairness, for the QoS equilibrium in BH. The mentioned ‘delay’ is particularly the packet sojourn time, which directly impacts the QoS. For example, in the satellite based remote meeting

service and the live broadcast service, the users situated in different cells should be provided with almost the same level of delay experience. However, the traditional allocation algorithm including the capacity fairness algorithm, hasn’t considered the delay fairness, hereby resulting in unbalanced QoS. To accomplish such delay fairness through resource allocation is just the work we consider here. The delay fairness problem has been mentioned in [7], but no time slot allocation algorithm is given there. Intuitively, processing all cells’ packets in the first-come-first-serve (FCFS) mode could achieve ideal delay fairness, but the beam hopping which is a physical technology involving power amplifying and time synchronization cannot act so fast like the packet scheduling technology [16]. As far as we know, there isn’t any precedent particularly on the algorithm of delay fairness for beam hopping. As supplement, the job we have done in this paper is to design the per-period hopping strategy to gain the equilibrated QoS for the system without changing the anterior capacity allocation.

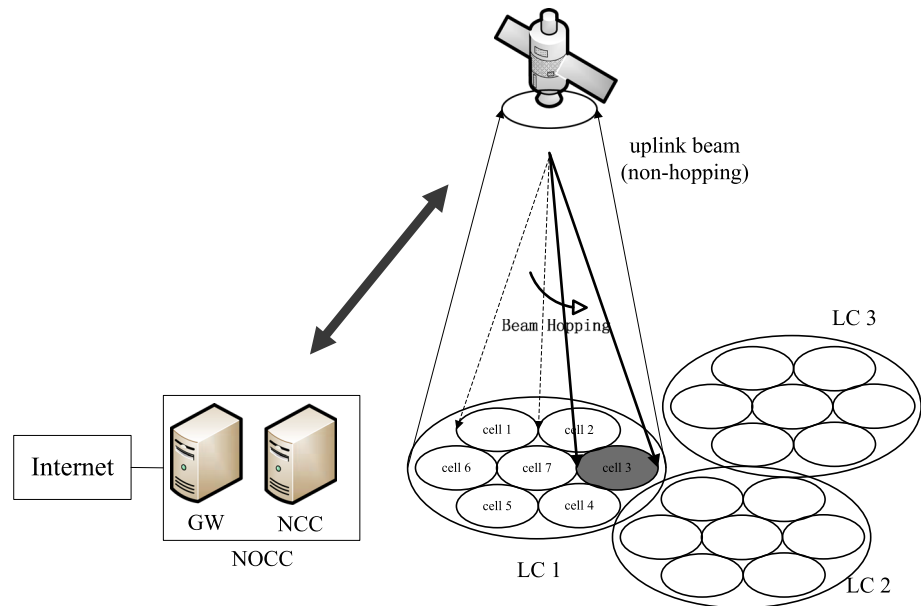
In this paper, we consider a general case for research of delay fairness: the cells in coverage have different traffic intensity, different channel attenuation and different dynamic instantaneous backlog. The time resource is allocated in per-period basis by using the queuing theory. After modeling with the target of delay proportional fairness (PF) among all the cells, we find it a long-term optimization problem and also on-line decision problem. In order to solve the stochastic programming problem, we introduce the time sharing principle and propose a beam hopping policy based on the stochastic gradient theory (SGT). The explicit expression of visiting time duration for each cell as well as the hopping sequence is obtained at the beginning of each beam hopping window (BHW). In addition, we research the convergence, the superiority, the robustness and the capacity region under fairness in simulations.

The rest of this paper is organized as follows: Sect. 2 introduces the beam hopping based satellite communication system, and the delay fairness problem is formulated in Sect. 3. In Sect. 4, we analysis and transform the problem using the SGT, and propose the delay fairness oriented beam hopping (DFBH) approach. From the point of view of the fairness and throughput, we compare the performance of proposed algorithm with other three existing beam hopping algorithms by simulations in Sect. 5. Finally, Sect. 6 concludes our work.

2 System model

2.1 Broadband satellite communication system

The considered satellite communication system is shown in Fig. 1, referring to RSM-A air interface protocol. Network

Fig. 1 Beam hopping model

Control Central (NCC) manages the access and allocates bandwidth to User Terminals (UTs) that are distributed sparsely in the coverage region. Gateway (GW) is in charge of the traffic from and to the internet. The two entities are situated in Network Operation Control Central (NOCC). The satellite in research is installed with multiple hopping spot-beams, for uplink and downlink usages. Every uplink (non-hopping) spot-beam serves a large extended coverage, called Large Cell (LC) in this paper. One LC is further divided into M cells (here, $M = 7$) with equal size, all of which are illuminated by one single downlink hopping beam through an appropriately designed illumination pattern. That is to say, one LC is served by two beams, one for hopping downlink and the other one for non-hopping uplink. The hopping beams are generated by beam forming or smart antenna technology, which works in a wide frequency band so as to provide broadband downlink service for the UTs. In addition, each hopping beam also transforms to large-cell beam for periodical signaling intention, consistent to the Spaceway-3 system.

Taking LC 1 as an example, we illustrate the detailed procedures for the system as follows.

1. Each UT runs its Bandwidth Request Estimation (BRE) algorithm to calculate its bandwidth request. Then each UT in LC 1 access the reverse link beam in multiple access mode (in Spaceway-3 system, the mode is MF-TDMA) to apply a certain service (such as stream media service) from the far internet and to report its channel attenuation to NOCC. This action is executed every hopping decision period.
2. The NCC receives the traffic demands of all the UTs, and then allocates appropriate bandwidth resource (in

TDM mode in Spaceway-3 system) to them according to Dynamic Bandwidth Allocation (DBA) scheme. After interaction, forward data transmission begins.

3. The GW buffers all the traffics of all UTs, and queues them again on the basis of their destinations, i.e. one queue for one cell (packet scheduling is also performed in this step). Beam hopping-associated scheduler (in the NOCC) multiplexes all the cells' queues into one downlink TDM stream based on the hopping pattern (generated by the beam hopping approach discussed in this paper).
4. The hopping beam in charge of LC 1 broadcasts (in form of large-cell beam) the slot synchronization signaling and hopping pattern information to all the UTs for notification in LC 1. Then it starts to transmit signals to the corresponding cells slot-by-slot, according to the destination of each slot until the period is end.

Explanation All the requested services of cells are satisfied with enough bandwidth by the DBA in NCC, which would not be affected by the beam hopping discussed here. However, the resulting different queues to deliver, which have diversity queue lengths, arrival rate and departure rate, are the very concern of the beam hopping. Therefore, although the aim of the beam hopping is to guarantee the fairness among the queues, the guarantee of traffic throughput is in its precondition.

2.2 Beam hopping scheme

The beam hopping techniques provide the system with greater flexibility in accommodating irregular and time variant traffic requests across the coverage region,

slot arrangement to obtain a *time-spatial transmission plan* in each BHW. Let $\delta_{i,n}$ denote the choice of arranging the i th cell to serve in the n th slot ('1' is yes, '0' is no), and then the plan could be expressed as follows.

$$plan = \begin{bmatrix} \delta_{1,1} & \cdots & \delta_{1,NUM} \\ \cdot & \delta_{i,n} & \cdot \\ \delta_{M,1} & \cdots & \delta_{M,NUM} \end{bmatrix}. \quad (3)$$

There are M^{NUM} possible solutions, making the brute search approach NP-hard. To make things worse, the explicit relation expression between the slot arrangement and the average packet sojourn time is precisely unreachable. If a cell could be served all the time, then the queuing model is $M/D/1/\infty$, and if the cell is not served all the time, then the queuing model is $M/0/1/\infty$ (0 means that serving rate is zero). However, beam hopping makes the queuing process switching between $M/D/1/\infty$ and $M/0/1/\infty$ with uncertain transition probability. This is the unique challenge posed by the beam hopping.

In view of its hard formulation, we give some reasonable rules for the BH problem. First of all, the slots of serving the same cell are conjunct. Secondly, instead of using the slot number to model the serving time, we relax the integer constraint to consider it a continuous variable, denoting as visiting time duration for cell i in BHW n , $t_i(n)$, $i = 1, \dots, M$ ($\sum_{i=1}^M t_i(n) = T$). After it is calculated, the corresponding slot number could be obtained by $num_i(n) = \left\lceil \left[\frac{t_i(n)}{T} \times NUM \right] \right\rceil$, where $\lceil \cdot \rceil$ is the rounding procedure. Another parameter to optimize under this rule is the visiting sequence, denoting by $S(n)$ in BHW n . In addition, the backlog of cell i in BHW n is denoted as $Q_i(n)$ for further usage. We regard the average sojourn time as the delay performance and model the delay fairness problem in term of the PF [18], as follows.

$$\begin{aligned} \max_{S(n), t_i(n)} & \sum_{i=1}^M \log[1/E[w_i^{soj}]] \\ s.t. & \sum_{i=1}^M t_i(n) = T \end{aligned} \quad (4)$$

According to the queuing theory [14], the total sojourn time over all the packets in 'custom-minute', including the in-service time, is the integral of the instantaneous custom number multiplied by time. So the average sojourn time could be transformed into

$$\begin{aligned} E[w_i^{soj}] &= \lim_{t \rightarrow \infty} \frac{1}{Z_i(t)} \int_0^t N_i(t) dt \\ &= \lim_{n \rightarrow \infty} \frac{1}{Z_i(nT)} \sum_{n=1}^n \int_{(n-1)T}^{nT} N_i(t) dt. \end{aligned} \quad (5)$$

where $N_i(t)$ is the packet number in time instant t in the system and $Z_i(t)$ is the summed packet number arriving

into the cell till time instant t . Substituting (5) into (4), we get the problem rewritten as

$$\begin{aligned} \max_{S(n), t_i(n)} & \lim_{n \rightarrow \infty} f(\mathbf{W}(n)) \\ s.t. & \sum_{i=1}^M t_i(n) = T \end{aligned} \quad (6)$$

where $f(\cdot)$ is the efficiency function defined by $f(\mathbf{W}(n)) = \sum_{i=1}^M \log(1/W_i(n))$ and $W_i(n)$ denotes the average sojourn time after the BHW n , expressed as

$$\begin{aligned} W_i(n) &= \frac{1}{Z_i(nT)} \sum_{n=1}^n \int_{(n-1)T}^{nT} N_i(t) dt \\ &= \frac{Z_i((n-1)T) \cdot W_i(n-1) + \int_{(n-1)T}^{nT} N_i(t) dt}{Z_i(nT)}. \end{aligned} \quad (7)$$

The objective of the scheduling is a long-term fairness while the action of the scheduling is a short-term slot arrangement (visiting time duration and the visiting sequence in every BHW). Therefore, this is an on-line decision problem. The sequence is an integer variable and the visiting time duration is a continuous variable, and moreover they are coupled to influence the overall delay: given a sequence, there is always a group of optimal visiting time durations to maximize the fairness; and vice versa. To solve the problem, we will decouple them in the next section.

Here, we list some easily confused variables in Table 1 which will be used in the whole context.

4 Delay fairness oriented beam hopping

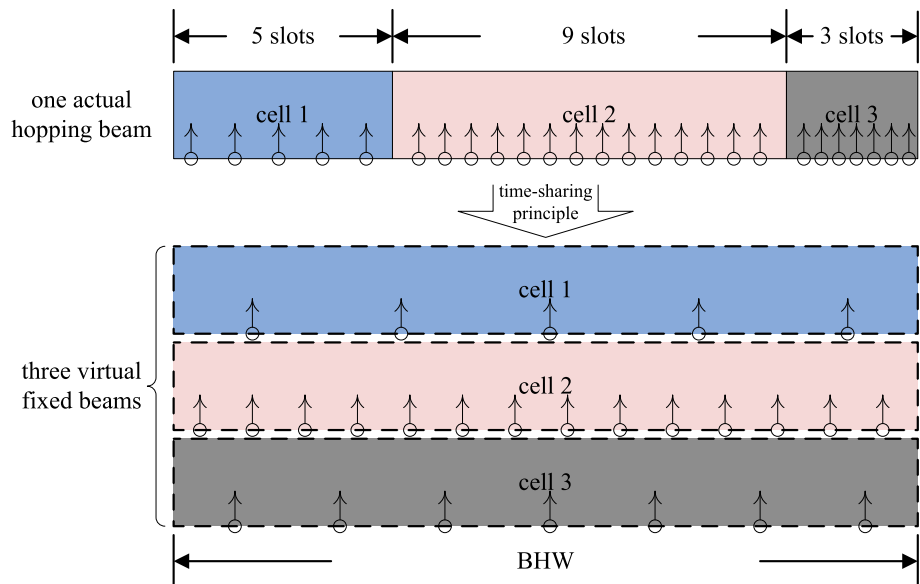
In this section, we will propose our approach for the delay fairness problem, namely DFBH. The breakthrough point behind the approach is to decouple the visiting time duration and the visiting sequence by the time-sharing principle.

4.1 Time-sharing principle

The basic idea behind the time-sharing principle is to relax the constraint of sequential illumination for all the cells. To explain the principle, we firstly describe an equivalent circumstance of the beam hopping. The considered hopping beam could be seemed as M virtual fixed beams with each one illuminating one cell, and the fixed beams are not turned on all the time, i.e. every beam could only illuminate its corresponding cell in the cell's visiting time duration. Using the time-sharing principle, for each beam-cell pair, we disassemble and then uniformly distribute the visiting time duration over one whole BHW in one virtual fixed beam, with its summed value unchanged, shown in Fig. 3. With the help of this principle, there is no visiting

Table 1 Some easily confused variables

Notation	Description	Notation	Description
$t_i(n)$	Visiting time duration for cell i in BHW n	w_i^{soj}	Sojourn time variable
num_i	Visiting slot number for cell i in BHW n	$W_i(n)$	Average sojourn time after the BHW n

Fig. 3 An illustration of the equivalent deterministic process for packet departure

sequence anymore, so that it could be ignored temporarily. Though it is possible that more than one cell may be illuminated simultaneously under this principle, we would compensate this drawback in the latter designation of visiting sequence. Note that this disposal only takes effect in the current (short-term) BHW to facilitate the analysis, while it cannot be used in long-term duration (otherwise, it would become capacity allocation).

The similar time-sharing principle has also been used in the context of subcarrier assignment in multiuser OFDM systems, such as the work [19]. While its function is to convert a mixed integer programming problem into a convex optimization problem, in this paper its function is to convert a sequential optimization problem into a parallel optimization problem.

Therefore, given the result of the visiting time durations, the serving process for cell i , which switches between u_i -deterministic process (illuminated) and non-process (not illuminated), is expanded to be an equivalent deterministic process. The mean value v_i and the variance σ_i^2 of the packet departure interval of the equivalent deterministic process are given correspondingly as follows.

$$\begin{cases} v_i = \mu_i \times t_i(n)/T \\ \sigma_i^2 = 0 \end{cases} \quad (8)$$

4.2 Optimization of visiting time duration via stochastic gradient theory

According to the SGT [20], in order to obtain the long-term optimization of the efficiency function $f(\cdot)$, we just need to maximize its increment between two conjunct BHWs, i.e. $\Delta f(\mathbf{W}(n))$.

We firstly deduce the partial derivative $\frac{\partial f}{\partial W_i(n)} = -1/W_i(n)$, and then the gradient of the efficiency function can be given as

$$\begin{aligned} \nabla f(\mathbf{W}(n)) &= \left[\frac{\partial f}{\partial W_1(n)}, \frac{\partial f}{\partial W_2(n)}, \dots, \frac{\partial f}{\partial W_M(n)} \right] \\ &= \left[-\frac{1}{W_1(n)}, -\frac{1}{W_2(n)}, \dots, -\frac{1}{W_M(n)} \right]. \end{aligned} \quad (9)$$

Therefore, we obtain the increment as follows.

$$\begin{aligned} \Delta f(\mathbf{W}(n)) &= f(\mathbf{W}(n)) - f(\mathbf{W}(n-1)) \\ &\approx \nabla f(\mathbf{W}(n-1)) \cdot [\mathbf{W}(n) - \mathbf{W}(n-1)] \\ &= -\sum_{i=1}^M \frac{W_i(n) - W_i(n-1)}{W_i(n-1)}. \end{aligned} \quad (10)$$

Using the expected value instead of the instantaneous value in (7), we can obtain

$$W_i(n) = \frac{(n-1)T\lambda_i W_i(n-1)}{nT\lambda_i} + \frac{[Q_i(n-1) + T\lambda_i] \times \mathbf{E}[w_i^{soj}]_n}{nT\lambda_i} \quad (11)$$

where $\mathbf{E}[w_i^{soj}]_n$ is the expected sojourn time for cell i in BHW n .

In the visiting time duration of cell i , The packet queuing process in one BHW could be approximately treated as $M/G/1/\infty$ based on the time-sharing principle, with its serving (packet departure) interval following deterministic distribution:

$$\begin{cases} v_i = \mu_i \times t_i(n)/T \\ \sigma_i^2 = 0 \end{cases} \quad (12)$$

Therefore, we can derive the expected sojourn time [14] in BHW n :

$$\begin{aligned} \mathbf{E}[w_i^{soj}]_n &= \mathbf{E}[w_i^{que}]_n + \mathbf{E}[w_i^{ser}]_n \\ &= \frac{\lambda_i + \lambda_i \sigma_i^2 v_i^2}{2(v_i^2 - \lambda_i v_i)} + \frac{1}{v_i} \\ &= \frac{2t_i(n)/T \times \mu_i - \lambda_i}{2t_i(n)/T \times \mu_i \times (t_i(n)/T \times \mu_i - \lambda_i)} \\ &= \frac{1}{2t_i(n)/T \times \mu_i} + \frac{1}{2(t_i(n)/T \times \mu_i - \lambda_i)}. \end{aligned} \quad (13)$$

Substituting (13) into (11), and then substituting (11) into (10), we obtain the following equivalent optimization problem.

$$\min_{t_i} \sum_{i=1}^M \frac{[Q_i(n-1) + T\lambda_i]}{T\lambda_i W_i(n-1)} \left[\frac{1}{2t_i/T \times \mu_i} + \frac{1}{2(t_i/T \times \mu_i - \lambda_i)} \right]. \quad (14)$$

where t_i is the simplification of $t_i(n)$ in BHW n . In order to gain the fairness, we just need to manage the optimization in every BHW. Noticing that the function is actually convex, it could be solved by some iterative optimizations such as the Newton's method [21]. Though this kind of approach could converge to its optimal solution by proper algorithm design, it needs spend lots of resource of computing, which adds extra load on the resource-limited satellite system. Therefore, it is more appealing to obtain the close-form solution of this problem, mainly from the implementation perspective.

Using the Lagrange multiplier method [21], we introduce a Lagrange multiplier β to combine the constrained expression, obtaining the Lagrangian as follows.

$$\begin{aligned} \mathcal{L}(\{t_i/T\}, \eta) &= \sum_{i=1}^M A_i \times \left[\frac{1}{2t_i/T \times \mu_i} + \frac{1}{2(t_i/T \times \mu_i - \lambda_i)} \right] \\ &\quad + \beta \times \left(\sum_{i=1}^M \frac{t_i(n)}{T} - 1 \right), \end{aligned} \quad (15)$$

where $A_i = \frac{[Q_i(n-1) + T\lambda_i]}{T\lambda_i W_i(n-1)}$. Differentiating the Lagrangian (15) with respect to $\frac{t_i}{T}$, we obtain

$$\begin{aligned} \frac{\partial \mathcal{L}(\{t_i/T\}, \beta)}{\partial t_i/T} &= \frac{-A_i}{2\mu_i} \times \frac{1}{(t_i/T)^2} + \frac{-A_i}{2} \\ &\quad \times \frac{\mu_i}{(t_i/T \times \mu_i - \lambda_i)^2} + \beta. \end{aligned} \quad (16)$$

Let $\frac{\partial \mathcal{L}(\{t_i/T\}, \beta)}{\partial t_i/T} = 0$, and we obtain

$$\frac{1}{(t_i/T)^2} + \frac{1}{(t_i/T - \lambda_i/\mu_i)^2} = \beta \times \frac{2\mu_i}{A_i}. \quad (17)$$

Since the equation involves high order variables, it will be complex for the solution $t_i/T = f(\beta)$, where the function f represents the solution with respect to β . So it will be intractable when solving the substituted equation, i.e. $\sum_{i=1}^M t_i/T = \sum_{i=1}^M f(\beta) = 1$, to obtain the exact value of β .

Therefore, we give up pursuing the optimal close-form solution, but attempt to attain a sub-optimal close-form solution. Before our going on, we give the following theorem and its corresponding lemma to assist our work.

Theorem 1 Defining the reference function $\mathcal{G}(\vec{t}/T) = \sum_{i=1}^M \frac{A_i}{t_i/T \times \mu_i}$ and the candidate variable set $\mathfrak{R} = \{(t_1, t_2, \dots, t_M) | \sum_{i=1}^M t_i/T = 1\}$, we have the conclusion that $\min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T) < \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p})$, where $\vec{p} = (\frac{\lambda_1}{\mu_1}, \frac{\lambda_2}{\mu_2}, \dots, \frac{\lambda_M}{\mu_M})$ and $\vec{t} = (t_1, t_2, \dots, t_M)$.

Proof The proof is given in “Appendix 1”.

Lemma 1 The following inequality expression (18) holds.

$$\begin{aligned} \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T) &< \min_{\vec{t} \in \mathfrak{R}} \frac{\mathcal{G}(\vec{t}/T) + \mathcal{G}(\vec{t}/T - \vec{p})}{2} \\ &< \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p}). \end{aligned} \quad (18)$$

Proof The proof is given in “Appendix 2”.

Defining the function $\ell(\vec{p}) = \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p})$, we can learn from Lemma 1 that the minimal value of our objective function has an upper bound $\ell(\vec{p})$ and a lower bound $\ell(0)$. In addition, recalling of the Theorem 1, we could conclude that the partial derivative of function $\ell(\vec{p})$ is positive, which says the function is monotone increasing with respect to each p_i . So we can draw the conclusion: there exists such a ratio η ($0 < \eta < 1$) that the minimal value of our objective function is equal to $\ell(\eta \cdot \vec{p})$. This conclusion suggests that we could equivalently obtain our target solution $opt = (t_1^{opt}, t_2^{opt}, \dots, t_M^{opt})$ by the following equation.

$$opt = \arg \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \eta \cdot \vec{p}). \quad (19)$$

This equation could be easily solved by Lagrange Multiplier Method, as follows.

$$\frac{t_i^{opt}}{T} = \frac{\eta \cdot \lambda_i}{\mu_i} + \frac{\varepsilon_i}{\sum \varepsilon_i} \left(1 - \sum \frac{\eta \cdot \lambda_i}{\mu_i} \right). \quad (20)$$

The solution could be explained from the following implementation perspective: first, allocate the slot resources by the constant ratio η , i.e. $\frac{\eta \cdot \lambda_i}{\mu_i}$; then, allocate the residual time sources according to the proportion of ε_i , i.e. $\frac{\varepsilon_i}{\sum \varepsilon_i} \left(1 - \sum \frac{\eta \cdot \lambda_i}{\mu_i} \right)$. In this sense, we name the constant allocation ratio η the *Guaranteed Ratio* (GR). Note that in different traffic intensity, the optimal GR value is different. Though the optimal GR value could not be given explicitly, a sub-optimal solution could be obtained by choosing a value from its value set $\{\eta | 0 \leq \eta \leq 1\}$. The principle for choosing the GR value is proposed as follows.

1. When the system traffic intensity is extremely high, i.e. $\sum \frac{\lambda_i}{\mu_i} = 1$, it is set to be 1.
2. When the system traffic intensity is extremely low, i.e. $\sum \frac{\lambda_i}{\mu_i} \approx 0$, it is set to be 0.
3. When the system traffic intensity is medium, i.e. $0 < \sum \frac{\lambda_i}{\mu_i} < 1$, it is set to be $\eta = \sum \frac{\lambda_i}{\mu_i}$.

It should be noted that $W_i(n-1)$ is needed in the calculation of A_i . However, in the ratio $\frac{\varepsilon_i}{\sum \varepsilon_i}$, it could be removed when the system is in long-term stable fairness, i.e. $W_i(n-1) = W_j(n-1), \forall i, j$. So far, the result of visiting time duration in the scheduling problem is determined by three parameters: the arrival rate λ_i , the serving rate μ_i and the instantaneous backlog $Q_i(n-1)$. That is to say, the fairness orient slot allocation depends not only on the traffic intensity and channel attenuation, but also on the instantaneous backlog.

In addition, the allocated slot number for each cell is calculated by

$$num_i = \left\lceil \left[\frac{t_i(n)}{T} \times NUM \right] \right\rceil. \quad (21)$$

4.3 Designation of the visiting sequence

The deduced visiting time duration in each BHW is based on the time-sharing principle, which actually ignores the hopping sequence. In this subsection, considering it is impossible to compute the optimal sequence for each BHW in polynomial time, we propose an instinctive hopping sequence instead. And this disposal is also adopted by the work [12], which chooses the cell with the longest backlog to illuminate in every slot. Our proposed sequence $S(n)$ in

BHW n is determined by the backlog length $Q_i(n-1)$ at the end of BHW $n-1$, i.e.

$$S(n) = (s_1, s_2, \dots, s_M) | Q_{s_1}(n-1) \geq Q_{s_2}(n-1) \dots \geq Q_{s_M}(n-1). \quad (22)$$

where the symbol $a|b$ means the variable a must satisfy the condition b . And s_i denotes the cell in the i th place of the visiting sequence. The $Q_i(n-1)$ could be obtained by statistics as shown in Fig. 2.

After obtaining the hopping sequence as well as the slot allocation, we could construct the entire beam-hopping pattern.

At last, we make a conclusion, particularly on the course of beam hopping approach, as the compensation to the step (3) in Sect. 2.1.

Sub-step 1 In the beginning of period n , obtain the channel attenuation α_i and look up the backlog info $Q_i(n-1)$ from each cell, and estimate the packet arrival rate λ_i , and the achievable channel capacity C_i .

Sub-step 2 Using these parameters, NCC calculates the visiting sequence using (22) and the allocated visiting slot number for each cell using (21). Then construct the beam hopping pattern.

5 Simulation and analysis

5.1 Simulation setup

Simulation setup: one GEO satellite is operated in Ka band with transparent transponder. In our concerned region, one uplink non-hopping beam is working with 30 MHz bandwidth. $M = 4$ in a LC, and the downlink hopping spot-beam is working with bandwidth $B = 100$ MHz. The power is fixed for the hopping beam $P = 100$ W. The BHW length is $T = 40$ ms and total slot number in one BHW is $NUM = 100$ (slot length is 400 us). We carry out Monte Carlo simulation by 1000 runs in the total simulation time 4 s. Packet arrival rates are randomly generated in the stability constraint $\sum_{i=1}^M \lambda_i / \mu_i \leq 1$ for all cells and keep unchanged throughout the simulation. Packet's arrival and departure are generated following the Poisson process with rates claimed in respective subsection next. And the channel attenuation α_i is influenced by the transmit distance and the weather condition, which is generated in empirical value region $[-160, -140]$ dB in simulations. Since the value of the channel attenuation is very small in satellite communications, we adopt the channel-to-noise ratio (CNR) [19] to render this parameter in simulation, expressed as $CNR = \frac{\alpha_i P}{BN_0}$. The noise temperature at the earth station is set to be 140 K.

The fairness and the throughput performances are evaluated among four beam hopping approaches.

1. DFBH proposed in this paper.
2. Jiang's Capacity Fairness based Beam Hopping (namely CFBH) in [9]. In CFBH, the slot arrangement is as follows.

$$\frac{\text{num}_i}{\text{NUM}} = \frac{\lambda_i/\mu_i}{\sum_{i=1}^M \lambda_i/\mu_i}. \quad (23)$$

3. Neely's 'Choose-the-K-Largest-Connected-Queue' method (namely CKLCQ) in [12]: in every slot, the cell with the longest backlog is chosen to serve.
4. Choi's 'Highest Attenuation-Weighted Demand First' method (namely HAWDF) in [13]. Its chosen cell in slot n should satisfy:

$$\text{cell}(n) = \arg \max_i (\alpha_i \times Q_i(n)). \quad (24)$$

To measure the fairness, we introduce the Jain coefficient [22] as follows.

$$J(W_1, W_2, \dots, W_M) = \frac{(\sum_{i=1}^M W_i)^2}{M \sum_{i=1}^M W_i^2}. \quad (25)$$

The larger J means a fairer performance.

5.2 The convergence and superiority

In this subsection, we research the convergence and superiority of the four algorithms. In order to observe their performances used in heterogeneous cells, we devise a typical circumstance for the toy simulation. For the cell one to three, the same CNR value is set but with different traffic arrival rate. For the cell three and four, the same traffic arrival rate is set but with different CNR values. Details are shown in Table 2. We can get the GR value $\eta = \sum \frac{\lambda_i}{\mu_i} = 0.84$.

Figure 4 shows the convergence of the four algorithms. After some perturbation, all the algorithms reach their stable states. It can be found that the proposed method significantly outperforms CKLCQ and HAWDF, i.e. almost 0.99 comparing to 0.92 and 0.87. This is mainly due to the reason that they two are not delay-oriented, i.e. CKLCQ is toward the stabilization while HAWDF considers the throughput optimization. CKLCQ always gazes at the packet queuing length and puts more service to the cell with the longest backlog length, therefore losing the delay fairness. HAWDF makes some improvement with more considerations of the channel attenuation, but it ignores the influence of the traffic intensity. Our method tailors the visiting time duration of the target beam for each cell by fully considering three related parameters and

designs beam-hopping sequence for each BHW. Comparing to our method, CFBH aims at the capacity fairness while overlooks the information revealed by the backlog, so it could not obtain a good delay fairness. In addition, the convergence rate of CFBH is faster than the CKLCQ and HAWDF.

In Fig. 5, we put our sight into the average packet sojourn time of each cell. When the channel attenuations of different cells are same, the resource allocation depends on the traffic arrival rate, i.e. the higher arrival rate the more resource allocated, no matter the allocation algorithm is throughput-oriented or fairness-oriented. Observing cell 1, cell 2 and cell 3 in CKLCQ, HAWDF and CFBH, an interesting result can be obtained that, when the traffic arrival rate is high, its packet sojourn time is not large but small, relatively. This can be explained as follows: CKLCQ and HAWDF that make their allocation decisions mainly on the backlog info, would consequentially serve the cell with low arrival rate only after its backlog reaches a high level; CFBH has the ability to balance the packet arrival rate and the packet departure rate, but cannot balance the packet sojourn time among all the cells. When the traffic arrival rates of different cells are same, the resource allocation depends on the channel attenuations. Learned from cell 3 and cell 4 in CKLCQ, a good channel attenuation would lead to a high packet serving rate, such reducing the average sojourn time. Learned from cell 3 and cell 4 in HAWDF and CFBH, the good channel attenuation would also release the pressure of always serving, resulting in a longer average sojourn time. These seem incompatible, but actually rational from the perspective of different allocation targets. In our method, we target at the delay fairness, balancing the influences of channel attenuation and the traffic arrival rate by the indicator of packet backlog. Therefore, our delay fairness performance exhibits better than the other three ones. In addition, as well as we know, the delay fairness and the delay efficiency are conflicting to some extent. This aspect could be seen from this figure that the average sojourn time among all cells in our DFBH is a little higher than that in other algorithms.

5.3 The robustness

Since fairness is more applied in heterogeneous cases, in this section, we investigate the algorithm robustness from

Table 2 Two groups of parameters in the toy simulation

Cell	1	2	3	4
Traffic arrival rate (/ms)	5	10	15	15
CNR	4e−3	4e−3	4e−3	6e−3

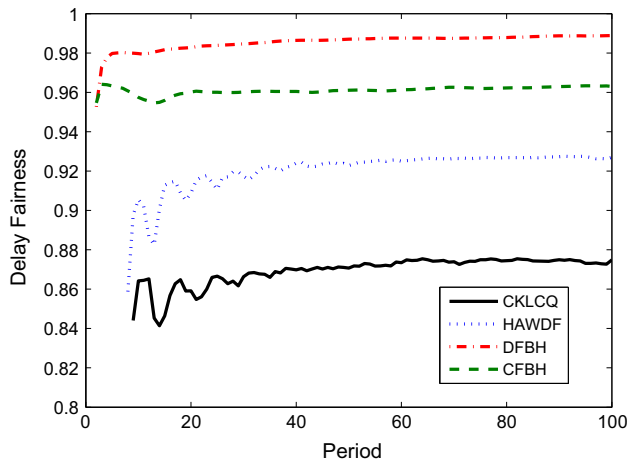


Fig. 4 Delay fairness comparison

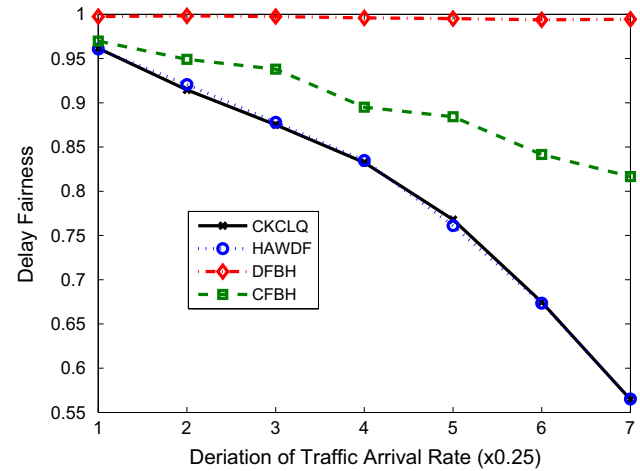


Fig. 6 Influence of deviation of arrival rates

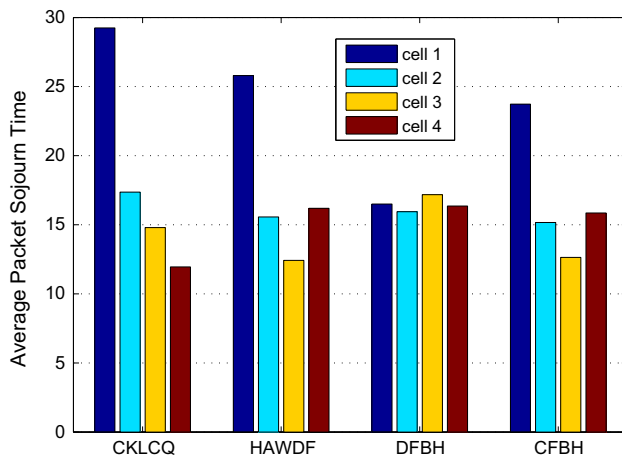


Fig. 5 The average sojourn time of each cell

the perspective of delay fairness when the arrival rates and the channel attenuations are heterogeneous.

In Fig. 6, we assume the channel attenuations of all cells are same and set different traffic arrival rate by varying their deviation. Numerically, $CNR = 0.001$ for all cells, and the vector of arrival rate is $\lambda = [1.75 - \sigma_\lambda, 2.00 - \sigma_\lambda, 3.25 + \sigma_\lambda, 3.50 + \sigma_\lambda]/\text{ms}$, where the deviation σ_λ increases from 0 to 1.5 by the step 0.25. We can conclude that the curve of delay fairness for each algorithm is descending as the deviation is larger, because the diversity of the heterogeneous system is more and more serious. That is to say, all scheduling methods have their own ability limitations to keep the delay fairness. In comparison, DFBH maintains the highest fairness above 0.98 while the CFBH has fairness above 0.8. The worst ones are the CKLCQ and HAWDF, as they descend steepest. This is mainly due to the non-consideration of the fairness in their original designation. In addition, since the channel attenuations are equal, the ‘attenuation-weighted’ approach,

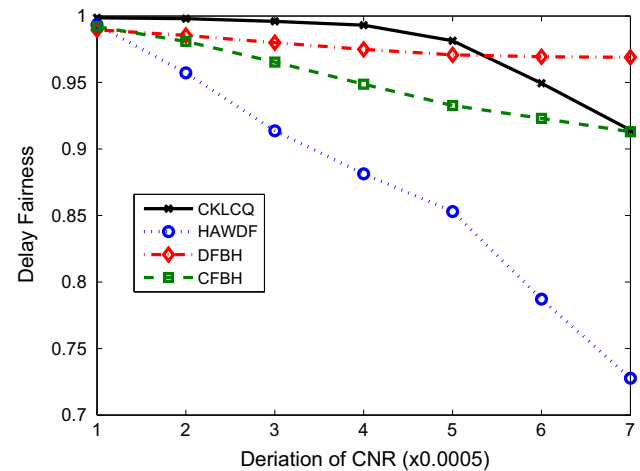


Fig. 7 Influence of deviation of channel attenuations (the same arrival rate)

i.e. HAWDF, is converted into CKLCQ, so they two have almost the same performance as shown in figure.

In Figs. 7 and 8, we research the influence of deviation of channel attenuations based on same and different traffic arrival rates correspondingly. In Fig. 7, $\lambda_i = 0.7/\text{ms}, \forall i$, and we set the vector of CNRs for all cells as $\overrightarrow{CNR} = [5.0 \times 10^{-3} - \sigma_{cnr}, 5.5 \times 10^{-3} - \sigma_{cnr}, 8.0 \times 10^{-3} + \sigma_{cnr}, 8.5 \times 10^{-3} + \sigma_{cnr}]$, where the deviation σ_{cnr} increases from 0 to 0.003 by the step 0.0005. We can see that the delay fairness is descending as the deviation of channel attenuations increases. But the performance of our proposed DFBH deteriorates the least, which verifies its advantage. A phenomenon can be noticed that when the deviation is small, CKLCQ gains better delay fairness in this setup. This can be explained by the backlog info: when the traffic arrival rate is the same, the backlog reveals the packet departure rates, so choosing the cell with the longest backlog is coincident with the delay fairness principle. The

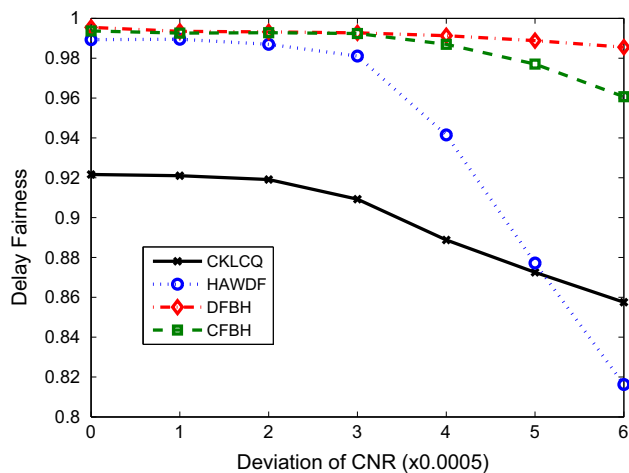


Fig. 8 Influence of deviation of channel attenuations (different arrival rate)

proposed DFBH performs a little worse (its fairness is still higher than 0.97) than the CKLCQ, mainly due to the approximation in its deduction. As the deviation is large, the advantage of DFBH is still very obvious.

More general cases are that the traffic arrival rates are heterogeneous while the channel attenuations are also heterogeneous. In Fig. 8, the setups are the same as those in Fig. 7, except the vector of traffic arrival rates is $\lambda = [5, 5, 9, 9]/\text{ms}$. As the deviation increases, the delay fairness of each algorithm is descending. The CKLCQ is not robust as it loses much delay fairness in this case comparing with the last case, while our DFBH still keeps the delay fairness of above 0.97.

5.4 The capacity region under fairness

In this subsection, we investigate the relation between the capacity and fairness.

Note that our proposed algorithm for delay fairness is based on the stability condition [23], i.e. $\sum_{i=1}^M \lambda_i / \mu_i \leq 1$. That is to say, although our algorithm could equilibrate the packet delay performance for all the cells, it doesn't sacrifice (e.g. block up) the arrival traffic. We define the capacity region under the stability constraint as

$$C_s = \left\{ (\lambda_1, \lambda_2, \dots, \lambda_M) \mid \sum_{i=1}^M \lambda_i / \mu_i \leq 1 \right\}. \quad (26)$$

However, for each vector of the traffic arrival rate in the region C_s , the obtained delay fairness after using our proposed resource allocation algorithm would be different. This argument has been verified by the results of the simulations in the above two subsections. However, from the view of the cell, we wonder how much traffic could be sustained while satisfying a given fairness criteria?

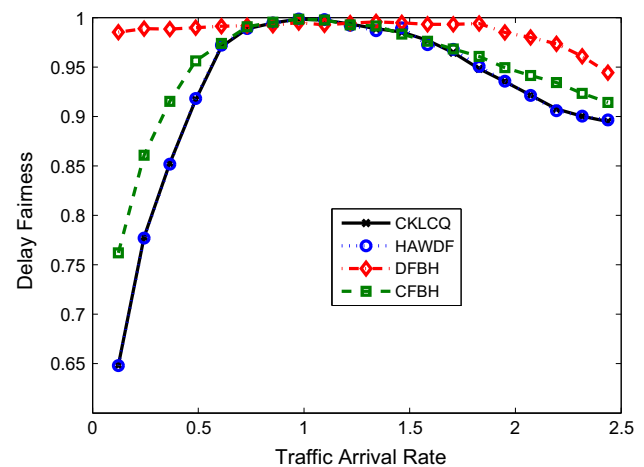


Fig. 9 Delay fairness versus traffic arrival rate

A simple simulation has been performed, with the special circumstance parameters set as: $\lambda = [\lambda, \lambda, 2, 2]/\text{ms}$ and $\text{CNR} = [0.001, 0.001, 0.001, 0.001]$. From the expression (26), we know the feasible value of the variable λ has the constraint: $\lambda \leq \mu \times \left(1 - \sum_{i=3}^4 \lambda_i / \mu_i\right) / 2 = 4.875$. Figure 9 shows the obtained delay fairness for all the possible λ , using the four algorithms respectively. When the variable λ is too small or too large, the delay fairness is relatively low because of the large diversity. And when its value is round 2/ms, the delay fairness could be higher than 0.95. Different algorithms have different concave curves. Our DFBH performs the best as it keeps the delay fairness always in a high level. This phenomenon inspires us to conceive the capacity region under fairness, as follows.

$$C_f = \{(\lambda_1, \lambda_2, \dots, \lambda_M) \mid J(W_1, W_2, \dots, W_M) \geq J_{th}\}. \quad (27)$$

where J_{th} is the endurable fairness threshold. The function $J(W_1, W_2, \dots, W_M)$ has been defined in expression (25) and W_i has been defined in expression (11).

More explicitly, we draw the capacity region under fairness for each algorithm, as show in grey in Fig. 10. The channel conditions are fixed at $\text{CNR} = 0.001$ and later $\text{CNR} = 0.002$ for all cells. $J_{th} = 0.97$. Other simulation parameters are the same as those in last simulation. From the comparison, our DFBH has the longest capacity region under the fairness threshold comparing with the other three algorithms. This is coincident with the simulations in above subsections. When CNR increases from 0.001 to 0.002, the packet departure rate increases, so the absolute values of the packet sojourn time become smaller. Therefore, the gap of packet sojourn time becomes more obvious, making the delay fairness worse. This is the reason why the capacity regions of CKLCQ and HAWDF are shorten when the CNR increases. Different from the throughput-oriented algorithms, our DFBH will spare more time slots to smooth

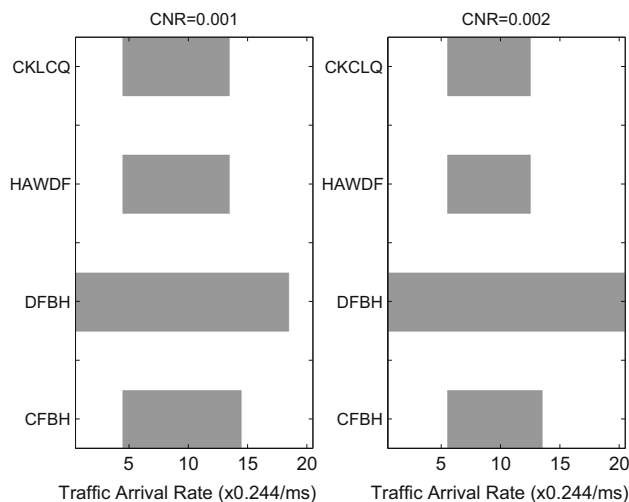


Fig. 10 The capacity region under fairness

the difference exposed by the increasing CNR , so its capacity region has become longer as expected.

To sum up, we list all the findings in our simulations as follows.

1. Capacity fairness is not equivalent to the delay fairness;
2. Under the throughput maximizing target, the cell with large traffic reversely enjoys low delay fairness;
3. Delay fairness conflicts with delay efficiency;
4. The larger is the deviation, the lower is the delay fairness, no matter it is the traffic arrival rate deviation or the channel attenuation deviation.

6 Conclusion

In this paper, we discuss the delay fairness problem encountered by cells that share one hopping beam in broadband satellite communication system. Due to the physical limitation, the beam couldn't hop as fast as the packet scheduling, making the ideal FCFS strategy unrealizable. Based on the condition of capacity satisfaction, we propose to dynamically schedule the slot number for each cell so as to realize the long-term delay fairness. We give out a complete beam hopping model as well as the BHW model. In order to solve the long-term optimization problem with its parameters coupled with each other, we introduce the time-sharing principle to help ease the complexity, and apply the SGT to deduce the close-form solutions of the visiting time duration. Combined with our proposed beam-hopping sequence, the entire hopping pattern is obtained. Our simulation results prove that the proposed algorithm has a significant performance gain over existing works in

the fairness performance. In our future work, multi-beam hopping (scheduling) combined with the other resource allocations will be further discussed.

Acknowledgments This work was supported by the National Natural Science Foundation of China (Grant No. 61231011). The authors are grateful to the anonymous referee for a careful checking of the details and for helpful comments that improved this paper.

Appendix 1: Proof of Theorem 1

Without loss of generality, we define $\vec{t}^* = (t_1^*, t_2^*, \dots, t_M^*)$ as the optimal solution for minimizing the reference function under constraint, i.e.

$$\vec{t}^* = \arg \min \mathcal{G}(\vec{t}/T). \quad (28)$$

subjecting to $\sum_{i=1}^M t_i^*/T = 1$. Since $\vec{t}^*$ is the minimum value, for any $\vec{t} \in \mathfrak{R}$, we have that

$$\mathcal{G}(\vec{t}/T) \geq \mathcal{G}(\vec{t}^*/T). \quad (29)$$

Substituting this $\vec{t}^*$ into the shifted function, we have that

$$\mathcal{G}(\vec{t}/T - \vec{p}) > \mathcal{G}(\vec{t}^*/T). \quad (30)$$

Therefore, we have

$$\mathcal{G}(\vec{t}/T - \vec{p}) > \mathcal{G}(\vec{t}^*/T). \quad (31)$$

for any given $\vec{t} \in \mathfrak{R}$. This is to say,

$$\min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p}) > \mathcal{G}(\vec{t}^*/T). \quad (32)$$

This completes the proof.

Appendix 2: Proof of Lemma 1

At first, it can be easily known that

$$\begin{aligned} & \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T) + \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p}) \\ & \leq \min_{\vec{t} \in \mathfrak{R}} [\mathcal{G}(\vec{t}/T) + \mathcal{G}(\vec{t}/T - \vec{p})]. \end{aligned} \quad (33)$$

Combining with the Theorem 1 which says $\min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T) < \min_{\vec{t} \in \mathfrak{R}} \mathcal{G}(\vec{t}/T - \vec{p})$, we know

$$\min_{\vec{t} \in \mathfrak{R}} 2\mathcal{G}(\vec{t}/T) \leq \min_{\vec{t} \in \mathfrak{R}} [\mathcal{G}(\vec{t}/T) + \mathcal{G}(\vec{t}/T - \vec{p})], \quad (34)$$

which proves the left part of (18). Then, we define a new function $\varphi(\vec{t}/T) = \sum_{i=1}^M \frac{A_i}{t_i/T \times \mu_i} + \sum_{i=M+1}^{2M} \frac{A_i}{t_i/T \times \mu_i - \lambda_i}$ with

$2M$ variables, i.e. $\vec{t} = (t_1, t_2, \dots, t_M, t_{M+1}, \dots, t_{2M})$, where $\mu_i = \mu_{i+M}, A_i = A_{i+M}, \lambda_i = \lambda_{i+M}$.

In addition, the candidate variable vector set is set as $\mathfrak{R}' = \{(t_1, t_2, \dots, t_{2M}) | (t_1, t_2, \dots, t_M) \in \mathfrak{R}, t_i = t_{i+M}, \forall i\}$. Learned from the proof of the Theorem 1, we know that the conclusion of Theorem 1 could also be used to the new function, i.e.

$$\min_{\vec{t} \in \mathfrak{R}'} \varphi(\vec{t}/T) < \min_{\vec{t} \in \mathfrak{R}'} \varphi(\vec{t}/T - \vec{q}),$$

$$\text{where } \vec{q} = (\underbrace{\bar{p}, 0, 0, \dots, 0}_M).$$
(35)

This proves the right part of the (18):

$$\min_{\vec{t} \in \mathfrak{R}} \left[\mathcal{G}(\vec{t}/T) + \mathcal{G}(\vec{t}/T - \vec{p}) \right] < \min_{\vec{t} \in \mathfrak{R}} 2\mathcal{G}(\vec{t}/T - \vec{p}).$$
(36)

Till now, the proof is completed.

References

- Mokhtar, A., & Azizoglu, M. (2000). On the downlink throughput of a broadband LEO satellite network with hopping beams. *IEEE Communications Letters*, 4(12), 390–393.
- ESA Project: Beam Hopping Techniques for Multibeam Satellite Systems, <http://telecom.esa.int/telecom/www/object/index.cfm?fobjectid=29159>.
- Whitefield, D., Gopal, R., & Arnold, S. (2006). Spaceway now and in the future: On-board IP packet switching satellite communication network. In *Military Communications Conference, 2006. MILCOM 2006* (pp. 1–7). IEEE.
- Jacomb-Hood, A. W., Dentinger, A. M., & Maalouf, K. J. (2003). Apparatus, method, and computer program products for cell-hopping satellite communications. US Patent 6,522,643. 2003-2-18.
- Bever, M. E., Cooper, S. A., & Jue, R., et al. (2006). Downlink beam hopping waveform. US Patent 6,992,992. 2006-1-31.
- Reddy, N. (2002). Spot beam hopping packet schedule system. 2002-6-18.
- Pecorella, T., Fantacci, R., Lasagni, C., et al. (2007). Study and implementation of switching and beam-hopping techniques in satellites with on board processing. In *Satellite and space communications. 2007. IWSSC'07. International workshop on* (pp. 206–210). IEEE.
- Anzalchi, J., Couchman, A., & Gabellini, P., et al. (2010). Beam hopping in multi-beam broadband satellite systems: System simulation and performance comparison with non-hopped systems. In *Advanced satellite multimedia systems conference (ASMA) and the 11th signal processing for space communications workshop (SPSC), 2010 5th* (pp. 248–255). IEEE.
- Lei, J., & Viquez-Castro, M. (2011). Multibeam satellite frequency/time duality study and capacity optimization. *Journal of Communications and Networks*, 13(5), 472–480.
- Fonseca, N. J. G., & Sombrin, J. (2012). Multi-beam reflector antenna system combining beam hopping and size reduction of effectively used spots. *IEEE Antennas and Propagation Magazine*, 54(2), 88–99.
- Sharma, S. K., Chatzinotas, S., & Ottersten, B. (2014). Cognitive beamhopping for spectral coexistence of multibeam satellites. *International Journal of Satellite Communications and Networking*, 33(1), 69–91.
- Neely, M., Modiano, E., & Rohrs, C. (2002). Power and server allocation in a multi-beam satellite with time varying channels. In *IEEE INFOCOM* (Vol. 3).
- Choi, J. P., & Chan, V. W. S. (2005). Optimum power and beam allocation based on traffic demand and channel conditions over satellite downlinks. *IEEE Transactions on Communications*, 4(6), 2983–2993.
- Asmussen, S. (1987). *Applied probability and queues*. New York: Wiley.
- Xu, Y., Wang, J., & Wu, Q. (2011). Effective capacity region of two-user opportunistic spectrum access. *Science China Information Sciences*, 54(9), 1928–1937.
- Castro, M. A. V., & Granados, G. S. (2007). Cross-layer packet scheduler design of a multibeam broadband satellite system with adaptive coding and modulation. *IEEE Transactions on Wireless Communications*, 6(1), 248–258.
- ITU-T Recommendation G. 1010: End-user multimedia QoS categories, <http://www.itu-t.org>.
- Jalali, A., Padovani, R., & Pankaj, R. (2000). Data throughput of CDMA-HDR a high efficiency-high data rate personal communication wireless system. In *Proceedings of IEEE VTC*.
- Tao, M., Liang, Y.-C., & Zhang, F. (2008). Resource allocation for delay differentiated traffic in multiuser OFDM systems. *IEEE Transactions on Wireless Communications*, 7(6), 2190–2201.
- Kushner, H. J., & Yin, G. (2003). *Stochastic approximation algorithms and applications* (2nd ed.). Berlin: Springer.
- Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge, MA: Cambridge University Press.
- Xu, Y., Wang, J., Wu, Q., Anpalagan, A., & Yao, Y. (2012). Opportunistic spectrum access in unknown dynamic environment: A game-theoretic stochastic learning solution. *IEEE Transactions on Wireless Communications*, 11(4), 1380–1391.
- Neely, M. J., Modiano, E., & Rohrs, C. E. (2003). Power allocation and routing in multi-beam satellites with time varying channels. *IEEE Transactions on Networking*, 11, 138–152.



Han Han received his B.Sc. degree and M.S. degree from PLA University of Science and Technology, Nanjing, China in 2008 and 2011 respectively. He is currently pursuing his Ph.D degree in PLA University of Science and Technology. His current research interests include cognitive radio, satellite communication, resource allocation and game theory. He is the recipient of the Jiangsu Province Excellent Master Theses Award in 2012. He is also

the reviewer of several journals in his research area.



Xueqiang Zheng received his B.Sc. degree in Communications engineering and Ph.D degree in Communications engineering and information systems from Institute of Communications Engineering, PLA University of Science and Technology, Nanjing, China, in 2004 and 2009 respectively. He is a Lecture at PLA University of Science and Technology. His current research interests are cognitive radio, spectrum sensing and spectrum management.



Yuan Lin received her B.S. degree and M.S. degree from PLA University of Science and Technology, Nanjing, China in 2008 and 2011 respectively. She is currently pursuing her Ph.D degree in PLA University of Science and Technology. Her current research interests include satellite communication, MIMO and beamforming.



Qinfei Huang received the B.Sc. and Ph.D. degrees in communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2004 and 2010, respectively. He is currently with the Beijing Capital Highway Development Group Co. Ltd. His research interests are in the area of signal processing for wireless communications and intelligent transport systems.